

EDUCATION

University of Tehran, Institute of Geophysics	Tehran, Iran
<i>Master of Science, Geophysics (Exploration Seismology)</i>	2025 – 2027
Kharazmi University, Faculty of Earth Sciences	Tehran, Iran
<i>Bachelor of Science, Geology</i>	2021 – 2025
- Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025) - Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024) - Ranked 8th out of 60 students in the Undergraduate Studies	
University of Lille, International Academy Lille Hdf	Lille, France
<i>Water Resource Management (Hydro 2024) Summer School</i>	Summer 2024
- Completed 5 ECTS credits in water resource management topics. - Presented an oral defense on numerical groundwater modeling, focusing on changing water abstraction in the Weyib Watershed, Ethiopia.	

TEACHING EXPERIENCE

Kharazmi University	Tehran, Iran
<i>Programming Teaching Assistant</i>	April 2025 – June 2025
Programming Course, under supervision of Dr. Morteza Asemani	
Kharazmi University	Tehran, Iran
<i>Geostatistics Teaching Assistant</i>	February 2025 – June 2025
Geostatistics Course, under supervision of Dr. Ehsan Pegah	
Young Scholars Club	Tehran, Iran
<i>Instructor</i>	July 2024
Taught "Applied Data Science in Geology" for the Iran National IESO¹ 2024 Team	
Kharazmi University	Tehran, Iran
<i>Instructor</i>	April 2024 – June 2024
Organized and instructed the Advanced Python and Machine Learning Workshop	
NODET²	Tehran, Iran
<i>Teacher</i>	May 2021 – September 2021
Taught ITEO³ Marketing at Allameh Helli (Branches 5 & 8) and Farzanegan High Schools	

WORK EXPERIENCE

FEZtool	Tehran, Iran
<i>Co-Founder & Research and Development Specialist</i>	August 2023 – Present
Led Earth science research initiatives, developing advanced algorithms and conducting geospatial data analysis for scientific insights.	
International Astronomical Search Collaboration, NASA Partner	Remote
<i>Asteroid Search Team Member</i>	July 2025 – September 2025
Contributed to the Astronomy Enthusiasts of Iran 5 & 3 team in analyzing Pan-STARRS data, earning credit for 3 preliminary asteroid discoveries (Object IDs: P22c6kw, P12cZ6o and P12dtv1).	
Kharazmi University	Tehran, Iran
<i>Research Assistant</i>	October 2024 – January 2025
Performed MASW to extract dispersion images under supervision of Dr. Mehdi Talkhablou.	
ARYANIC	Tehran, Iran
<i>Project Manager</i>	November 2022 – June 2023
Managed a project titled "Business Factory", leading a team of 7	
GreenOly	Tehran, Iran
<i>Product Manager</i>	August 2022 – January 2023
Created an Entrepreneurship MOOC with over 15 Olympiad Medalists.	

¹International Earth Science Olympiad

²National Organization for Development of Exceptional Talents

³Iranian Thinking & Entrepreneurship Olympiad

COMMUNITY SERVICE

The 4th National Festival of Young Iranian Geologists

Education Committee Member

Tehran, Iran

June 2025 – July 2025

The 3rd National Festival of Young Iranian Geologists

Graphics Committee Member

Tehran, Iran

May 2024 – June 2024

Imam Khomeini International University

Editorial Board Member

Qazvin, Iran

September 2023 – June 2024

SELECTED PUBLICATIONS

Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. *Kharazmi Journal of Earth Sciences*.
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. *Ovan Scientific Journal*
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashefi, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. *Journal of Water Management Modeling (JWMM)*

Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In *Seismology Student Workshop (SSW 2025)*, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In *The 1st Conference on Applied Geophysics in Mines*, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In *2nd International Conference for Iranian Geography and Earth Sciences Students*, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In *8th Symposium of Sedimentological Society of Iran*, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In *26th Conference of Geological Society of Iran*, Urmia, Iran

Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence*. **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning*. **Tasir Publication**.

PROJECTS

PlanetaryGeophysics | github.com/aradfarahani/PlanetaryGeophysics *2025 – Present*

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.
- Created a 3D interactive Mars globe and mapped gravity, seismic, and magnetic data.

FEZrs (Open Source) | pypi.org/project/fezrs *2023 – Present*

- Advanced Python library for remote sensing, developed by FEZtool with over 80,000 downloads

Seismicity Analysis in Iran (2010–2025) | iris.edu/hq/workshops/2025/06 *2025*

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

Goelectricspy (Open Source) | github.com/aradfarahani/Goelectricspy *2023*

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

LICENSES & CERTIFICATIONS

Seismology Skill Building Workshop 2025, EarthScope Consortium	2025
The Science of the Solar System, California Institute of Technology (Offered by Coursera)	
AstroTech, The University of Edinburgh (Offered by Coursera)	
Space Exploration, Technical University of Munich (Offered by Coursera)	
Rocket Science for Everyone, Yale University (Offered by Coursera)	
International Data Masterclass, United Nations Big Data	2024
Application of Remote Sensing in Geology, Zawinas	
Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method, (NIGS) ¹	
Global Environmental Management, DTU ² (Offered by Coursera)	2023
The 2 nd Summer School of Geology, University of Tabriz	
Remote Pilot Certification (Multirotor), CAAIRI ³	
Unmanned Aerial Vehicle Surveying, Maan Academy	
GIS, Mapping, and Spatial Analysis Specialization, University of Toronto (Offered by Coursera)	
Hydraulic Modeling of flood plains in Urban Rivers, Kharazmi University	2022
CS50 Certificate, CS50 (Harvard University)	2021

HONORS & AWARDS

LeetCode Global Ranking - <i>Ranked Top 100</i> , LeetCode	2025
Deep-ML Global Leaderboard - <i>Ranked Top 50</i> , Deep-ML.com	
Master's Entrance Exam (Geophysics) - <i>Ranked 8th Nationwide</i>	
Hydro 2024 Research-Oriented Summer School - <i>Full Scholarship</i> , University of Lille	2024
Geological Society Logo Design Contest - <i>Winner</i> , Kharazmi University	
Research Visit Grant (2500 DKK) - <i>Awardee</i> , University of Copenhagen	2022
DrCT 2021 Global Computational Competition - <i>Gold Medalist</i>	2021
1 st national Thinking & Entrepreneurship Olympiad (ITEO) - <i>Silver Medalist</i>	2020
WMI 2015 World Mathematical Championships Selection - <i>Silver Medalist</i>	2015

SKILLS & INTERESTS

Programming Languages: Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R

Markup Languages: Markdown, L^AT_EX

Softwares: Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas

Field Techniques: Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition

Languages: Farsi (Persian): *Native*; English: *Fluent (C1)*

Research Interests: Signal Processing, Applied Geophysics, Remote Sensing, Planetary Science, Machine Learning, Image Processing, Open-Source Software Development, Inverse Problems

PROFESSIONAL MEMBERSHIPS

Space Generation Advisory Council	Member, 2025 – Present
Google Maps Platform Innovators	Member, 2025 – Present
Geological Society of Iran	Student member, 2021 – 2022

¹National Iranian Geophysical Society

²Technical University of Denmark

³Civil Aviation Authority of Islamic Republic of Iran